GG-P-00661B (GSA-FSS)
AMENDMENT-2
January 30, 1973
SUPERSEDING
Amendment-1 (GSA-FSS)
March 16, 1972

AMENDMENT

TO

INTERIM FEDERAL SPECIFICATION

PROJECTOR, STILL PICTURE, OVERHEAD; ATTACHMENT AND ACCESSORIES

This amendment was developed by the Standardization Division, General Services Administration, Federal Supply Service, Washington, D. C. 20406, based upon currently available technical information. The General Services Administration has authorized this amendment as a part of Interim Federal Specification GG-P-00661B (GSA-FSS), dated February 8, 1971.

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Paragraph 1.2.1: Under classes delete in its entirety and substitute the following:

Class A - Auditorium Type Projectors (see table I):

Projectors designed for use in large rooms or auditoriums seating approximately 75 to 300 persons where maximum screen brightness is a major factor and portability is not critical.

Class B - Classroom Type Projectors (see table I):

Projectors designed for use in typical classrooms or meeting rooms seating approximately 15 to 75 persons.

Class C - Collapsible Office Type Projectors With Carrying Case (see table I):

Projectors designed for small room or office use, seating approximately 1 to 15 persons, where compact portability, low noise levels and short focal length are major factors.

Paragraph 2.1: Under "Federal Specifications" after the line starting with CC-M-636 add the following: "DD-G-451 - Glass, Plate, Sheet, Figured."

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Paragraph 2.1: Under "Military Specifications" delete the line "MIL-C-675 - Coatings of Glass Optical Elements". After the line starting with "MIL-I-983" add the following: "MIL-M-13508 - Mirror Coating Process, Front Surface Aluminized."

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Paragraph 2.2: Under "American National Standards Institute (ANSI), Inc. Standard"

Change the date from "Z1.4 - 1961" to "Z1.4 - 1971."

Paragraph 3.2: Delete the first sentence and substitute the following:

"Unless otherwise specified (see 6.2), before production is commenced, a finished unit covered by this specification and in accordance with only the ordering data (paragraph 6.2) specified in the "Invitation for Bid" shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.3.2.

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Paragraph 3.5: Delete in its entirety and substitute the following:

3.5 Design. The projector shall be equipped with a light source, internal or external reflector, cooling system, horizontal projection stage, optical system, and associated equipment such as a carrying case. The projector shall be requirements of table I.

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Paragraph 3.5: Table I. Insert a new column as follows:

Heading - Maximum Noise Level (3.20)
Unit line - Decibels
Class A line - 45
Class B line - 45
Class C line - 38

Paragraph 3.7.1, last sentence: Delete and substitute the following:

"The lamp shall be easily replaceable without the use of tools."

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Paragraph 3.11.2: Delete in its entirety.

Paragraph 3.11.3: Delete the last sentence and substitute the following:

"The cable jacket shall be either black, brown or gray."

Paragraph 3.11.4, line 2: Delete all after second sentence starting with "All other units are to have . . ."

Paragraph 3.11.5: Delete the first sentence and substitute the following:

"The cooling system shall be capable of cooling and ventilating the projector housing to meet the requirements of 3.19."

Paragraph 3.11.9: Delete the first sentence and substitute the following:

"A two-position or three-position switch for turning the projector "ON" and "OFF" or "ON", "OFF" and "BLOWER" shall be provided.

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Paragraph 3.12.1: Delete the last sentence in its entirety.

Paragraph 3.12.1.1: Delete in its entirety and substitute the following:

3.12.1.1 Projected image. The image produced by the projection lens shall be visually free from haze due to spherical aberration, internal barrel reflections, surface defects, glass defects and from both longitudinal and lateral chromatic aberration, when viewed at the projector using a projection distance of 10 feet. The allowable tolerance for one side of the zonal region shall be no more than 2 percent of the width of the projected picture area.

Paragraph 3.12.1.2: Delete in its entirety and substitute the following:

3.12.1.2: Distortion. The projected image of the straight lines formed by the sides of an 8 inch square placed centrally on the projection stage, shall not deviate from true straight lines by more than one percent of their length for classes A and B, or two percent of their length for class C projectors. (see 4.4.4)

Paragraph 3.12.2, last line: Delete "MIL-C-675" and substitute "MIL-M-13508."

Paragraph 3.12.3: Delete in its entirety and substitute the following:

3.12.3: External reflector, projection lamp. The projection lamp's external reflector, if used, shall be of first quality material and shall not crack, flake, peel, discolor or otherwise suffer impairment in quality of performance when subjected to the life test of 3.21 and the moisture test of 4.4.9.

Paragraph 3.12.5: Delete in its entirety.

Paragraph 3.12.7: Delete in its entirety and substitute the following:

3.12.7 Glare reducer. When specified (see 6.2), the projector shall contain a permanent glare reducing or low glare capability designed so that the glare to the eyes of any operator ranging in height from 5' -0" to 6' -4" and standing in either a right hand or left hand operating position shall be no more than 40,000 foot lamberts as measured in 4.5. Any glare

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reducing device or internal design shall be below the surface of the projector stage and shall not interfere with the operator in the performance of his duties.

Paragraph 3.13.1: Delete in its entirety and substitute the following:

Projection head. The projection head shall be firmly positioned and aligned with the optical axis. Any completely enclosed projection head, without provisions for cleaning shall be dust proof. Any non - dust proof head shall have a mirror and lenses easily removed or easily accessible for cleaning without the use of tools other than a screw driver. Lenses and mirrors shall be so constructed to insure the accurate return to their original positioning after removal for cleaning.

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Paragraph 3.13.2: Delete the second sentence and substitute the following:

When the projector head is connected directly to the vertical post by either forming or bending of the vertical post, the projection head support shall be of sufficient strength to support the projector without permanent deflection taking place when the projector is tested in accordance with 4.4.6.

Paragraph 3.13.3: Delete first sentence and substitute the following:

"Means shall be provided for raising the image a minimum of 20° above the horizontal."

Paragraph 3.13.4: Delete first sentence and substitute the following:

An easily operable rack and pinion, free from detrimental backlash, or cam, or a nonslip friction focusing mechanism, attached to the projection head support, shall be provided, which will produce a clear steady focus of the image in accordance with table I.

Paragraph 3.16: Delete in its entirety and substitute the following:

3.16 Projection stage. A projection stage approximately 10 inches by 10 inches in size. shall be provided. The stage shall provide a suitable surface for projecting written material, and for the projection of transparencies placed thereon either in a lengthwise or crosswise position. The fresnel lens or stage surfaces shall be readily accessible for cleaning without the use of tools other than a screwdriver.

Paragraph 3.16.1: Delete first sentence in its entirety and substitute the following:

When specified (see 6.2), the projector shall be equipped with two (2) roll attachments capable of accepting 50 feet of 10-1/2 ± 1/4 inch wide polyester or acetate operating from either top to bottom or side to side.

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Paragraph 3.19: Delete in its entirety and substitute the following:

3.19 Temperature rise. The temperature rise of the projector housing, after 2 hours of continuous operation shall not exceed 50°F. above an ambient temperature of 70°F. exclusive of the blower exhaust and the projection stage surface. The temperature rise of the test transparency after 5 minutes of operation shall not exceed 90 F. above an ambient temperature of 70°F. (see 4.4.15)

Paragraph 3.20: Delete in its entirety and substitute the following:

3.20 Noise level of cooling system. The noise level of the cooling system when measured in accordance with 4.4.10 shall not be greater than the decibels specified in Table I, above a reference level of 20 $\mu N/m^2$ in accordance with ANSI - S1.4 - 1971.

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Paragraph 3.2.5: Delete in its entirety and substitute the following:

3.2.5 Operational. When tested, as specified in 4.4.11, the projected image shall be in accordance with 3.12.1.1 and 3.12.1.3.

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Paragraph 4.3.2: Delete the first and second sentences and substitute the following:

"The responsibility for the inspection of the preproduction sample, as specified in 3.2, shall be in accordance with 4.1. The preproduction inspection shall be as specified in Tables II, III, IV and V except requirements omitted by ordering data (see 6.2).

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Paragraph 4.4.1: Delete "118 volt" and substitute "120 volt."

Paragraph 4.4.4, line 3: Delete the word "side" and substitute the word "line".

Paragraph 4.4.6: Delete in its entirety and substitute the following:

4.4.6 Support arm deflection. A clamp or similar holding device shall be securely fastened to the projection head support arm when the projector is assembled and ready for operation. Attach one end of a chain to the clamp and the other to a spring which is attached to a rigid support. The spring shall not allow the projector base to touch either the bench or floor when fully extended. Allowing the projector to hang free supported by the clamp, raise the projector by the base or sides five to seven inches, and release allowing the projector to drop free until stopped by the chain, spring and clamp. The spring shall not extend more than two to four inches before stopping ihe downward travel of the projector after release. repeat the drop six times. Check support arm for conformance with 3.13.1 and 3.13.2.

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Paragraph 4.4.9.1: Delete in its entirety and substitute the following:

4.4.9.1 Test as indicated on MIL-STD-170.

(a) Dry at 130°F for 4 hours.
(b) Condition at 86°F for 6 hours
(c) Cool at 68 F for 2 hours
(d) Subject to three continuous 24 hour, temperature changing cycles. Notes 1, 2, 3 and 4 on Mil-Std-170 shall apply except the temperature specified in (a) above.
(e) After cycling has been completed, condition the equipment for 24 hours at 75° ± 5°F and 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. The equipment shall pass all applicable operating tests in paragraph 4.4.

Paragraph 4.4.10, line 3: Change ANSI date from 1961 to 1971.

Paragraph 4.4.10, line 6: Change curve B to curve A.

Paragraph 4.4.12: Delete in its entirety and substitute the following:

4.4.12 Screen lumens output. To determine the luminous flux or screen lumens output, the projection equipment being tested shall be operated with a line voltage that is controlled to a tolerance of ± 0.5 percent of the rated voltage of the lamp being used, measured at the lamp terminals. The lamp(s) specified by the projector manufacturer shall be used for these tests. Measurement of the illuminance shall be made in a darkened room so that no more than 1/2 footcandle of ambient light strikes the screen. For those projectors which accept either horizontally or vertically oriented materials, a screen area four feet (1.2m) by four feet (1.2m) shall be used. (This area may be marked off on a larger surface if desired.) Any commercial illumination meter may be used as a light sensing instrument, provided the sensing element has a spectral sensitivity that matches the CIE (Commission International de L'Eclairage) luminosity curve for the CIE standard observer. The unit of measurement used shall be the footcandle.

NOTE: Luminous Flux is normally defined as:

Summation of readings in foot-candles times area of screen in sq. ft. divided by the number of readings. By specifying a screen of 16 square feet ($1.5m^2$) and using 16 readings, we then have a direct measure of luminous flux by merely adding the readings obtained.

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4.4.12.1 Measurement of Projector Equipment for Luminous Flux.

4.4.12.1.1 The 48" x 48" (1219.2mm x 1219.2mm) screen area shall be divided into 16 sections, one foot (.3m) by one foot (.3m). The direct (not reflected) illumination shall be read at the center of each section.

4.4.12.1.2 An appropriate mask shall be used to describe the proper projection aperture of the type of projection used, such as 2" x 2", (50.8mm x 50.8mm) 4" x 4", (101.6mm x 101.6mm) 10" x 10" (254mm x 254mm).

4.4.12.1.3 The projector shall be set so that its optical axis is perpendicular to aligned with the center of the target screen.

4.4.12.1.4 The projector shall be moved forward or back until the projected light just exactly fills the target screen when the focus has been adjusted for the sharpest aperture outline on the screen.

4.4.12.1.5 The light sensing device shall be positioned at and parallel to the plane of the viewing surface, to measure the projected light falling on the screen.

4.4.12.1.6 The illuminance at each of the 16 locations shall be measured and recorded.

4.4.12.1.7 The sum total of the 16 illumination measurements in foot-candles equals the light output in lumens.

Paragraph 4.14.14: Change paragraph number to 4.4.14.

Paragraph 4.14.15: Delete in its entirety and substitute the following:

4.4.15: Temperature rise. The equipment shall be operated and tested to determine compliance with the requirement specified in 3.19. The instrument used to determine the rise in temperature shall be accurate to $\pm$ 5°F. The temperature rise at the hottest spot of the projector housing shall be measured. A test transparency one inch square or larger and having a uniform optical density of 1.5 $\pm$ 0.1 shall be placed on the projection stage, at the center, on the optical axis. A temperature reading shall be taken on the bottom surface of the transparency at the end of five minutes for conformance with 3.19.

4.4.16: Delete in its entirety and substitute the following:

4.4.16 Uniformity of screen illumination. To measure the uniformity of screen illumination or light output of a projector, it is necessary to make a total of five measurements; one near each of the four corners (see 4.4.16.1) and one at the center of the area described in 4.4.12. Each of the four corner readings shall be expressed as percentage of the center reading. The uniformity of screen illumination shall be expressed as the average of the four corner over center reading percentages.

4.4.16.1 To locate the four corner positions, place the center of the light sensing device on the intersection of two lines drawn perpendicular to the horizontal and vertical sides of the screen area (4.4.12). Each perpendicular line shall be drawn 8" (63.50mm) from each corner in both directions.

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Paragraph 4.4.20: Delete in its entirety.

Paragraph 4.4.22C line 1: Delete 3 hours and substitute 1 1/2 hours.

Paragraph 4.4.22C line 2: Delete 30 minutes and substitute 15 minutes.

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